

Mohammad Toranjsimin

Ph.D. Applied Mathematics — M.Sc. Biostatistics

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Research Profile

Applied mathematician and biostatistician with expertise in statistical methodology, causal inference, machine learning, longitudinal data analysis, high-dimensional modeling, numerical analysis, optimization, and simulation studies. Experienced in developing theoretical methods, conducting Monte Carlo simulation studies, and analyzing complex health datasets using modern statistical and machine learning techniques. Strong mathematical background combined with applied research in biostatistics, statistical learning, and health data science.

Research Interests

- Statistical Methodology
- Causal Inference
- High-Dimensional Data Analysis
- Machine Learning
- Longitudinal Data Analysis
- Regularization Methods
- Optimization
- Numerical Analysis
- Scientific Computing
- Monte Carlo Simulation
- Biostatistics and Health Data Science

Education

University of Saskatchewan, Saskatoon, Canada

2023–2025

Master of Science (M.Sc.) in Biostatistics

Thesis: *Machine Learning for Causal Inference with Application in a Lung Health Study*

Supervisor: Prof. Li Xing

GPA: 91/100

Relevant Coursework:

- Computational Statistics
- Statistical Machine Learning in Data Science
- Mathematical Statistics and Inference
- Current Biostatistical Methods and Computer Applications
- Introduction to Epidemiology and Biostatistics

K. N. Toosi University of Technology, Tehran, Iran

2014–2019

Ph.D. in Applied Mathematics (Numerical Analysis)

Dissertation:

The Noncompact Integral Operators and their Applications to Numerical Solution of Singular Volterra Integral Equations

Supervisor: Prof. Mahmoud Hadizadeh Yazdi

GPA: 18.80/20 (Top 5%)

Iran University of Science and Technology, Tehran, Iran

2009–2011

M.Sc. in Applied Mathematics (Numerical Analysis)

Thesis:

Envelope Stepsize Control for Iterative Algorithms Based on Fejér Processes

Supervisor: Prof. Touraj Nikazad

University of Zanjan, Iran

2004–2008

B.Sc. in Applied Mathematics

Research Experience

Graduate Research Assistant

2023–2025

University of Saskatchewan

- Developed causal inference frameworks using propensity score methods, double machine learning (DML), and longitudinal mixed-effects models.
- Applied XGBoost and random forests to estimate nuisance parameters and treatment effects.
- Conducted Monte Carlo simulation studies evaluating bias, variance, coverage probability, and robustness of statistical estimators.
- Analyzed longitudinal clinical data from the Lung Health Study.
- Implemented reproducible workflows using R.
- Contributed to manuscripts, conference presentations, and knowledge dissemination.

Doctoral Research in Applied Mathematics

- Developed numerical methods for singular Volterra integral equations.
- Investigated theoretical properties of noncompact integral operators.
- Derived convergence and error analysis results.
- Applied spectral collocation and product integration methods.
- Studied applications to fractional differential equations.
- Implemented computational algorithms and numerical methods in scientific computing.

Applied Data Analysis and Statistical Research Projects

Machine Learning for Physical Activity Classification Using Accelerometer Data

(Supervisor: Daniel Fuller, University of Saskatchewan, Jan 2026 - present)

- Developed and evaluated machine learning models (Random Forest, Support Vector Machine, Neural Networks) to classify physical activity intensity using smartphone accelerometer data.
- Engineered 50+ time-domain and statistical features from raw triaxial sensor data to improve predictive performance.
- Implemented rigorous validation strategies, including train-test split and Leave-One-Participant-Out (LOPO) cross-validation, to assess model generalizability.
- Conducted hyperparameter tuning and model optimization using cross-validation techniques Visualized model performance using confusion matrix heatmaps and comprehensive evaluation metrics (Accuracy, F1-score, Sensitivity, Specificity)

Causal Inference and Longitudinal Analysis

(Supervisor: Li Xing, University of Saskatchewan, Sep 2023 - Sep 2025)

- Modeled longitudinal lung function decline using Linear Mixed Models.
- Applied propensity score matching, weighting, and regression adjustment.
- Implemented Double Machine Learning using XGBoost and random forests.
- Conducted simulation studies assessing treatment effect estimation.

Medical Cost Analysis

(Supervisor: Michael Szafron, University of Saskatchewan, Jan 2024 - Sep 2024)

- Investigated determinants of healthcare expenditures.
- Applied multiple linear regression and Weighted Least Squares.
- Conducted model diagnostics and validation.

Machine Learning for Land Classification

(Supervisor: Li Xing, University of Saskatchewan, Sep 2023 - Jan 2024)

- Applied K-means clustering, Fuzzy C-means clustering, and PCA.
- Analyzed satellite imagery and environmental data.
- Evaluated classification performance and temporal changes.

Peer-Reviewed Publications

1. Toranjsimin, M. and Hadizadeh, M. A priori mesh grading in collocation solution of noncompact Volterra integral equations with diagonal singularity. *International Journal of Computer Mathematics*, 2023.
2. Toranjsimin, M. and Hadizadeh, M. On the Class of Noncompact Volterra Integral Equations: Theory and Application to Fractional Differential Equations with Variable Coefficient. *Journal of Integral Equations and Applications*, 2020.

3. Toranjsimin, M. and Hadizadeh, M. Spectral Collocation Method for a Class of Integro-Differential Equations with Erdélyi-Kober Fractional Operator. *Advances in Applied Mathematics and Mechanics*, 2020.

Conference Presentations

1. Assessing the Effectiveness of Smoking Cessation Programs on Lung Function Decline: A Propensity Score and Mixed-Effects Model Approach.
Canadian Society for Epidemiology and Biostatistics (CSEB), Montreal, Aug 2025.
2. Comparing Double Machine Learning and Propensity Score Methods for Causal Inference in Lung Function Decline.
Statistical Society of Canada (SSC), University of Saskatchewan, May 2025.
3. Propensity Score and Double Machine Learning Approaches for Robust Causal Inference.
University of Saskatchewan Poster Day, October 2024.
4. Challenges in Regression Modeling: Inconsistency and Instability.
University of Saskatchewan Poster Day, October 2023.
5. A Trapezoidal Type Product Integration Method for a Noncompact Weakly Singular Volterra Integral Equation.
6th Seminar on Numerical Analysis and Applications, University of Maragheh, Maragheh, Iran, Sep 2016.
6. Reducing the Collocation Parameters for a Class of Weakly Singular Volterra Integral Equations.
47th Iranian Mathematics Conference, Kharazmi University, Tehran, Iran, Sep 2016.

Technical Skills

Programming

- R
- SAS
- MATLAB
- Mathematica
- Maple
- C/C++

Statistical Methods

- Causal Inference
- Propensity Score Methods
- Double Machine Learning
- Longitudinal Data Analysis
- Linear Mixed Models
- Generalized Linear Mixed Models
- High-Dimensional Data Analysis

- Machine Learning
- Monte Carlo Simulation

Mathematical Methods

- Numerical Analysis
- Optimization
- Integral Equations
- Differential Equations
- Scientific Computing

Reproducible Research

- Git
- GitHub
- LaTeX
- R Markdown

Teaching Experience

University of Saskatchewan 2023–2025

Teaching Assistant:

BIOS 990, MATH 224, MATH 226, MATH 133, MATH 163, MATH 110, MATH 101

K. N. Toosi University of Technology 2014–2018

Teaching Assistant:

Advanced Numerical Analysis, Numerical Linear Algebra, Ordinary Differential Equations (ODEs)

University of Zanjan 2005–2008

Teaching Assistant:

Differential Equations, Calculus I and II, Numerical Analysis

Islamic Azad University 2013–2017

Instructor:

Calculus I and II, Differential Equations, Engineering Mathematics, Linear Algebra, Statistics

University of Applied Science and Technology 2013–2022

Instructor:

Calculus I and II, Differential Equations, Linear Algebra, Statistics Operations Research I–III

Awards and Honors

- Ranked 3rd in the Regional Mathematical Olympiad of Iran (2007)
- Ranked 1st among all B.Sc. students, University of Zanjan (2008)
- Ranked 1st among all M.Sc. students, Iran University of Science and Technology (2011)

- Top 5% of Ph.D. students, K. N. Toosi University of Technology (2019)

International Research Experience

Visiting Researcher (Sabbatical)

Instituto Superior Técnico, Lisbon, Portugal

May 2018 – October 2018

Professional Service

- Seminar Coordinator, BIOS 990 Seminar Series, University of Saskatchewan, Saskatoon, Canada.
- Volunteer, Statistical Society of Canada Annual Meeting, University of Saskatchewan, Saskatoon, Canada, May 2025.

References

Dr. Daniel Fuller

Associate Professor

Department of Community Health and Epidemiology

University of Saskatchewan

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Dr. Li Xing

Associate Professor

Department of Mathematics and Statistics

University of Saskatchewan

Email: li.xing@math.usask.ca

Prof. Mahmoud Hadizadeh Yazdi

Professor

Department of Applied Mathematics

K. N. Toosi University of Technology

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Dr. Touraj Nikazad

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